ICC T20 World Cup 2024

Cricket Analytics

Subject: Data Analytics

Group Members:

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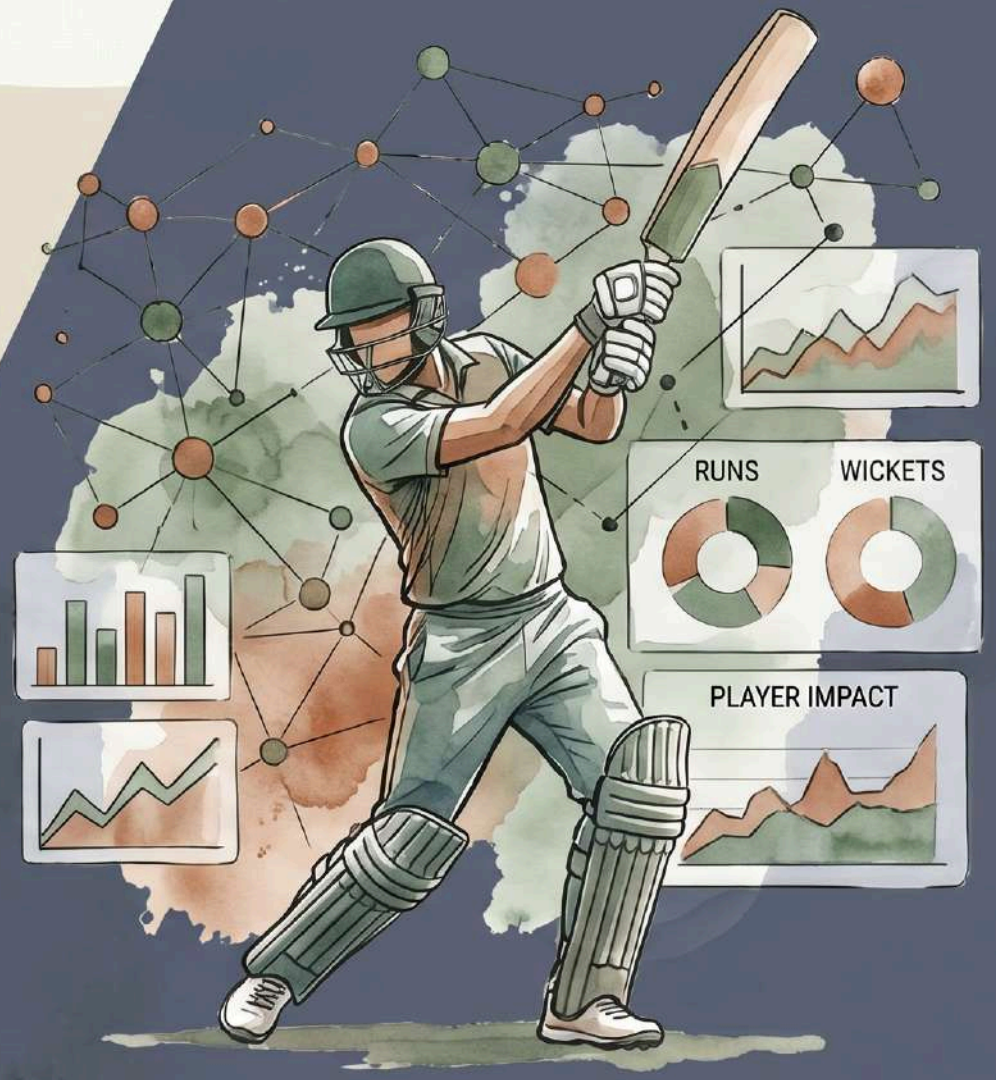
Introduction

Modern cricket generates vast, complex datasets.

Every delivery provides critical insights (runs, wickets, performance).

Data analytics converts raw data into meaningful intelligence.

This project analyzes performance trends in the ICC T20 World Cup 2024.



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Problem Statement

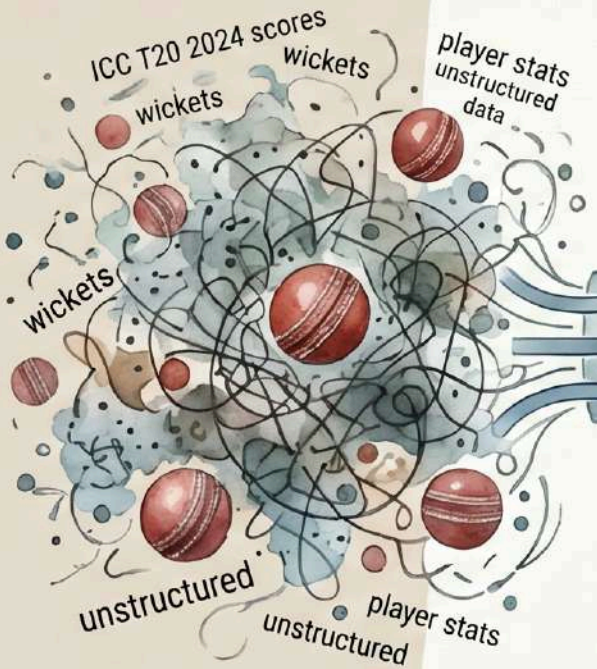
Cricket datasets are large, complex, and unstructured

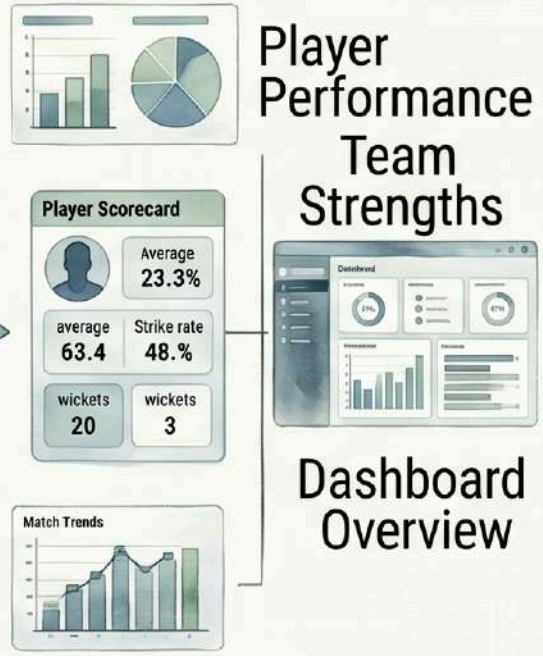
It is difficult to manually analyze such large volumes of data

Important insights like player performance, team strengths, and match trends are hidden

Lack of visualization makes it harder to interpret results

There is a need for a system that can simplify, analyze, and present the data effectively





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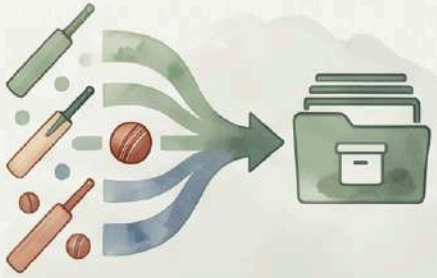
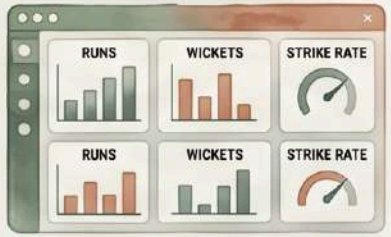
Objectives



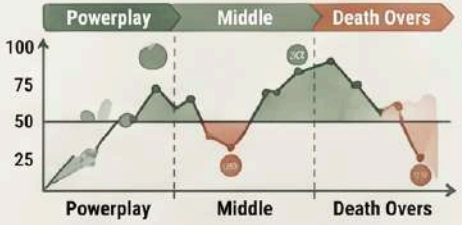
Collect/Organize

Clean/Preprocess

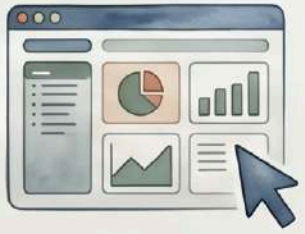


Identify KPIs

Analyze Batting/Bowling



Study Match Patterns



Create Dashboard

- To collect and organize cricket match data
- To clean and preprocess the dataset for accurate analysis
- To identify key performance indicators such as runs, wickets, and strike rate
- To analyze batting and bowling performance
- To study match patterns such as scoring trends in different phases
- To create a visually appealing and interactive dashboard

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Data Description

- Dataset: ICC T20 World Cup 2024 matches
- Type: Ball-by-ball dataset providing detailed match information
- Dataset includes:
  - Match details (teams, date, venue)
  - Batting data (runs scored, strike rate)
  - Bowling data (wickets, economy rate)
  - Extras such as wides and no-balls
  - Match phases (Powerplay, Middle overs, Death overs)
- This detailed dataset allows in-depth performance analysis

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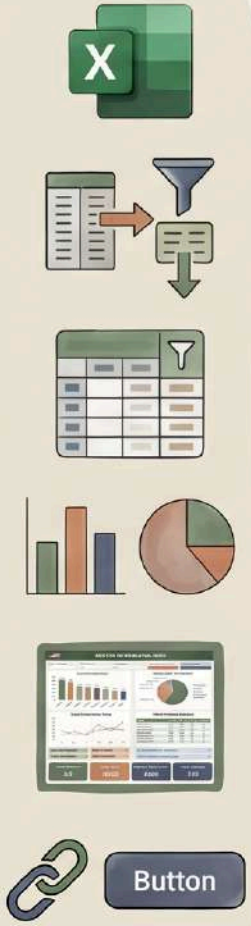
Tools & Techniques Used

- Microsoft Excel used as the primary tool for analysis
- Data cleaning performed to remove errors, blanks, and inconsistencies
- Pivot tables used to summarize and organize the data
- Charts such as bar charts and pie charts used for visualization
- Dashboard techniques used to present insights in an interactive format
- Hyperlinks and buttons added for easy navigation



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Key Performance Indicators (KPIs)

KPIs are important metrics used to measure performance. KPIs calculated in this project:

Total Runs Scored

13,064

Total Wickets Taken

684

Total Fours

968

Total Sixes

519

Number of Teams

20

Number of Venues

9

Average Runs per Match

251

Average Wickets per Match

13.2

These KPIs provide a quick overview of tournament performance

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Analysis Performed

Batting Analysis

- Identified top run scorers
- Compared player strike rates

Bowling Analysis

- Identified top wicket takers
- Analyzed economy rates

Phase Analysis

- Divided match into Powerplay, Middle, and Death overs
- Studied scoring patterns in each phase

Team Analysis

- Compared team performance across matches

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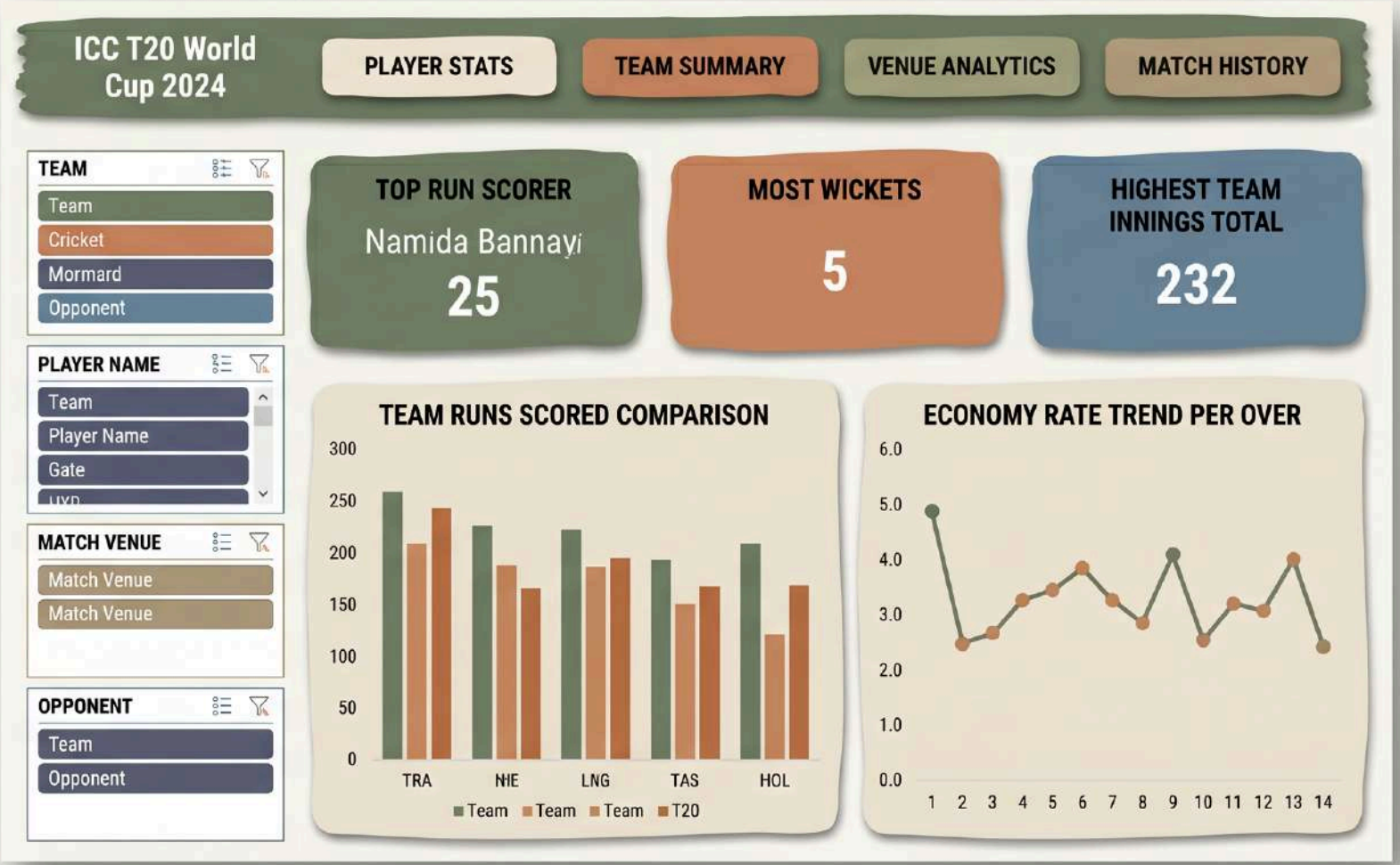
Dashboard Overview

- Comprehensive dashboard created using Excel
- Visual representation of key insights
- Dashboard includes:
  - KPI cards for quick metrics
  - Charts for performance comparison
  - Filters for better interaction
  - Navigation buttons for easy movement
  - Helps users quickly understand complex data



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Insights & Conclusion

Key Insights:

- Maximum runs are scored during the death overs
- Top-performing players have a major impact on match outcomes
- Teams with higher strike rates and lower economy rates perform better
- Field-first strategy is commonly preferred in matches

Conclusion:

- The project successfully transformed raw cricket data into meaningful insights
- It demonstrated the importance of data analytics in sports
- The dashboard provides a simple and interactive way to understand data
- This project enhanced skills in data analysis, visualization, and dashboard design

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